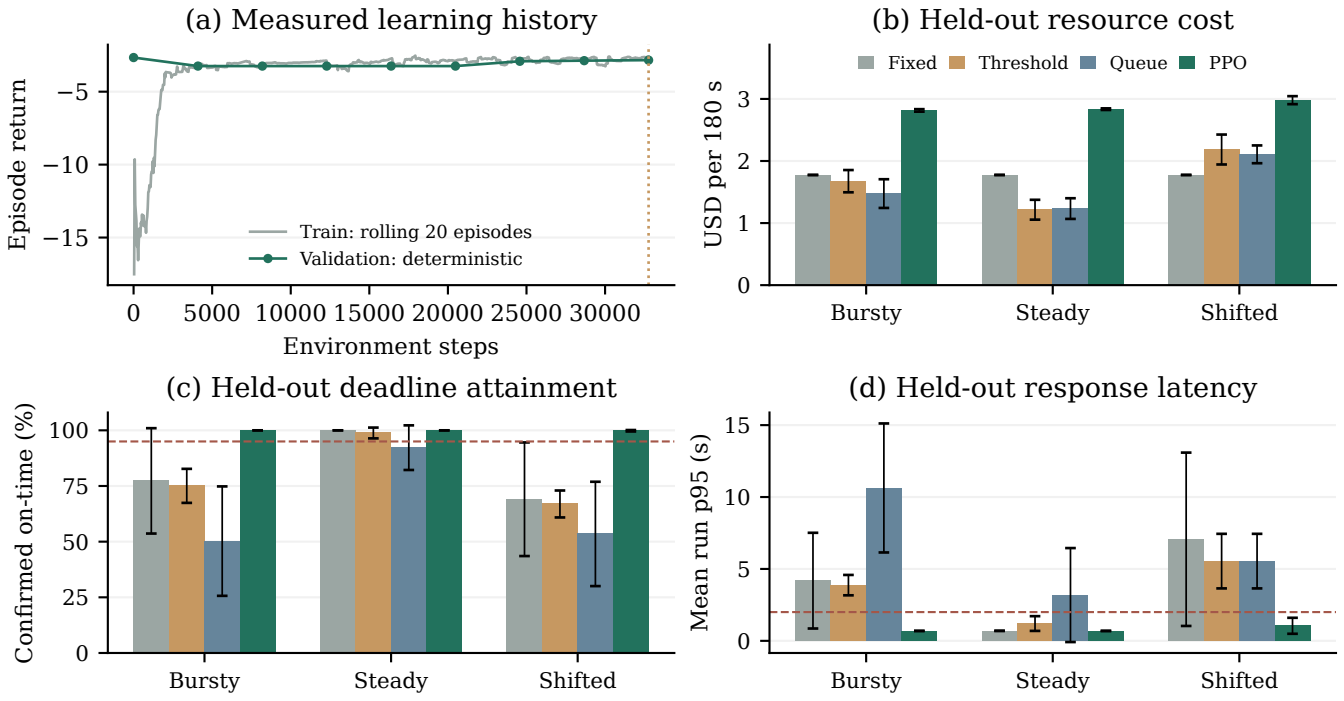
(a) Measured learning history
Episode return
−5
−10
−15
Train: rolling 20 episodes
Validation: deterministic
0
5000
10000
15000
20000
25000
30000
Environment steps
(b) Held-out resource cost
Fixed
Threshold
Queue
PPO
USD per 180 s
0
1
2
3
Bursty
Steady
Shifted
(c) Held-out deadline attainment
Confirmed on-time (%)
0
25
50
75
100
Bursty
Steady
Shifted
(d) Held-out response latency
Mean run p95 (s)
0
5
10
15
Bursty
Steady
Shifted